Enhancing Deepfake Detection through Dynamics of Facial Expressions

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***Abstract*—The rapid evolution of deepfake technology poses significant threats to digital security, media integrity, and public trust, necessitating the development of robust detection frameworks. Traditional deepfake detection methods primarily rely on pixel inconsistencies, frequency-domain analysis, or handcrafted features, but these approaches are increasingly vulnerable to advanced generative models that produce high-fidelity manipulations. In this study, we introduce MADDM, a Masked Autoencoder-based deepfake detection model that leverages facial expression dynamics to identify inconsistencies in muscle coordination—an aspect that remains challenging for deepfake generators to replicate accurately. Our model is trained in a self-supervised manner, first learning natural facial expressions from real datasets and then detecting anomalies in synthetic videos by reconstructing masked facial regions. Evaluations on Celeb-DF, DFDC, and FaceForensics++ datasets demonstrate that MADDM significantly outperforms existing detection methods, achieving an average accuracy of 81.1%, with state-of-the-art performance on Celeb-DF (86.3%). Further analysis through intra-dataset and cross-dataset testing confirms the model's superior generalization capabilities. The results highlight the potential of expression-based deepfake detection as a powerful and scalable solution for digital forensics and misinformation control. Future research should explore real-time implementation, transformer-based optimizations, and adversarial training strategies to enhance detection efficiency against evolving deepfake techniques.**

***Index Terms*—deepfake detection; facial expressions analysis; spatio temporal; convolutional neural network; masked auto encoders**

SECTION I - INTRODUCTION

The proliferation of deepfake technology has presented an unprecedented challenge in the realm of digital media authenticity. Deepfake videos, generated using deep learning models such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), can convincingly manipulate facial features, expressions, and even voice to create hyper-realistic synthetic content [1]. While deepfakes have applications in entertainment, accessibility, and creative industries, their misuse in misinformation campaigns, identity theft, and cyber fraud necessitates the development of robust detection mechanisms [2].

Traditional deepfake detection techniques primarily focus on pixel-level inconsistencies, frequency domain artifacts, and frame-wise analysis [3]. However, recent advancements in generative models, such as StyleGAN and DeepFaceLab, have significantly reduced detectable visual artifacts, making conventional detection methods increasingly ineffective [4]. Given the human face's intricate structure, one potential avenue for improvement in deepfake detection is leveraging the natural dynamics of facial expressions, particularly the coordinated movement of facial muscles. Unlike synthetic facial animations, genuine human expressions adhere to well-established physiological constraints, such as the Duchenne marker, which reflects authentic emotional states through synchronized activity of the zygomatic major and orbicularis oculi muscles [5].

Deepfake generators, even the most advanced ones, struggle to maintain the natural synchrony between facial muscles when manipulating expressions. This phenomenon results in "disconnected facial expressions", where individual facial regions exhibit inconsistencies in movement. For instance, a synthetic smile may alter the curvature of the lips but fail to activate the surrounding orbital muscles, producing an unnatural visual cue. This deviation from expected muscle coordination provides a valuable forensic feature that can be leveraged for deepfake detection.

To analyze these discrepancies, we propose a Masked Autoencoder-based Deepfake Detection Model (MADDM), which learns the natural co-articulation of facial expressions and detects inconsistencies indicative of deepfake generation. Our approach builds upon self-supervised learning paradigms, utilizing a two-stage training process:

1. Facial Expression Encoding: The model is trained on real facial videos, where parts of the face are masked and reconstructed, forcing the system to learn natural muscle coordination.
2. Deepfake Discrepancy Detection: The pre-trained model is fine-tuned on deepfake datasets, identifying deviations from expected facial dynamics.

The remainder of this paper is structured as follows: Section II (Literature Review) discusses previous research on deepfake detection, emphasizing facial feature-based and AI-driven techniques. Section III divided in two parts, details the proposed approach, including Section III(a) for dataset selection, training pipeline, and Section III(b) for model architecture. Section IV (Results) presents experimental evaluations, comparing the MADDM model's performance against state-of-

the-art detectors. Section V (Discussion) analyzes the broader implications of our findings, potential adversarial attacks, and future research directions. Finally, Section VI (Conclusion) summarizes key insights and suggests future developments in AI-driven forensic detection.

SECTION II - LITERATURE REVIEW

Deepfake detection has gained significant attention in recent years due to the increasing sophistication of generative models and their potential misuse. This section reviews existing literature on deepfake detection methodologies, emphasizing facial expression dynamics as a forensic feature. We categorize prior work into three key areas: (1) pixel and artifact-based detection, (2) frequency-domain analysis, and (3) physiological and expression-based detection methods.

Pixel and Artifact-Based Detection

Early deepfake detection efforts focused on pixel-level inconsistencies and visual artifacts that generative models fail to correct. Li et al. (2018) introduced a convolutional neural network (CNN)-based approach that detects inconsistencies in skin texture and illumination, which deepfake models often fail to replicate accurately [6]. Following this, Spatio-Temporal Convolutional Neural Networks (ST-CNN) were introduced to enhance detection accuracy by capturing temporal dependencies in facial expressions, providing a more robust approach against frame-level inconsistencies [7]. Similarly, Rossler et al. (2019) developed FaceForensics++, a benchmark dataset containing real and manipulated videos, enabling CNN-based classifiers to identify generative artifacts such as blurred edges, unnatural skin tones, and misaligned facial contours [8].

Despite their success, pixel-based techniques are vulnerable to adversarial attacks and post-processing enhancements, such as denoising or super-resolution, which remove detectable artifacts [9]. Furthermore, state-of-the-art GANs, such as StyleGAN3, incorporate adaptive instance normalization, which significantly reduces these pixel-level inconsistencies, making detection increasingly difficult [10].

Frequency-Domain Analysis for Deepfake Detection

To overcome limitations in pixel-space detection, researchers have explored frequency-domain analysis, leveraging the fact that generative models introduce statistical anomalies in high-frequency components. Durall et al. (2020) demonstrated that Fourier Transform-based methods effectively distinguish real and synthetic images, as deepfake videos tend to exhibit unnatural periodic patterns [11]. Moreover, Frank et al. (2020) utilized Discrete Cosine Transform (DCT) coefficients to detect discrepancies in local frequency components, particularly in facial regions that experience frequent movement, such as the eyes and mouth [12].

A mathematical basis for frequency-based detection stems from the power spectral density (PSD) of natural images, which follows a $1/f$ decay law [13]. Deepfake-generated images, however, deviate from this pattern due to GAN-induced interpolation artifacts. The frequency domain representation $F(u, v)$ of an image $I(x, y)$ is given by the 2D Fourier Transform:

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \tag{1}$$

where deviations from expected spectral decay indicate potential deepfake manipulation. While effective, frequency-based methods often struggle against deepfake models trained with adversarial loss functions, which increasingly align synthetic images with real-world spectral distributions [14].

Physiological and Expression-Based Detection

A more robust approach to deepfake detection focuses on human physiological constraints, such as gaze patterns, blinking rates, and facial muscle coordination. One of the earliest studies in this domain by Li et al. (2018) demonstrated that deepfake models fail to accurately reproduce natural blinking behavior, as training datasets frequently contain static portrait images rather than dynamic facial sequences [6]. However, once this limitation was publicized, deepfake generators adapted by incorporating eye movement synthesis, rendering blink-based detection less effective.

A more promising direction involves analyzing facial expression dynamics, which remain difficult for deepfake generators to synthesize accurately. The Duchenne smile, a classic example, requires coordinated activation of the zygomaticus major (cheek muscles) and orbicularis oculi (eye muscles), a feature that GANs often struggle to reproduce correctly [15]. Similarly, discrepancies in lip motion and eyebrow movement coordination serve as additional indicators of manipulation [16].

Our approach builds upon these findings by proposing a Masked Autoencoder (MAE)-based framework that learns the intrinsic dependencies between facial muscle activations. Unlike traditional deep learning classifiers that process entire images, MAEs mask random facial regions and reconstruct them based on learned spatial-temporal relationships, providing a self-supervised mechanism to detect expression inconsistencies. Inspired by He et al. (2022) [17] and [18], we employ a loss function that minimizes the reconstruction error between original and predicted facial expressions, defined as:

$$L_{MAE} = \sum_{i=1}^{N} ||E(F_i) - R(F_i)||^2 \tag{2}$$

where $E(F_i)$ represents the encoder-processed feature embeddings and $R(F_i)$ denotes the reconstructed facial expression vectors.

Existing deepfake detection methods, while effective in controlled scenarios, face limitations in generalizability and robustness against adversarial refinements. Pixel and artifact-based approaches are increasingly bypassed by advanced generative models, while frequency-domain techniques, though effective, require manual feature engineering and remain susceptible to noise. Physiological and expression-based detection

methods offer a promising alternative but require further refinement to account for natural human variability. Refer to Table I below for a detailed breakdown.

This paper aims to bridge this gap by introducing a facial expression dynamics-based deepfake detection model, leveraging masked autoencoder architectures to identify unnatural discrepancies in facial muscle coordination. Our approach enhances robustness by learning natural expression relationships through self-supervised learning, allowing for improved deepfake identification in real-world settings.

SECTION III(A) - DATASETS

To ensure a robust and comprehensive evaluation of our proposed deepfake detection model, we utilize multiple benchmark datasets containing both real and manipulated videos. Each dataset provides unique challenges, allowing us to assess the generalizability of our approach against different types of deepfake manipulations.

The FaceForensics++ dataset [8] is one of the most widely used datasets for deepfake research, containing high-quality manipulated videos generated using FaceSwap, DeepFake, and NeuralTextures. These videos cover various compression levels, making them suitable for testing detection robustness under different levels of quality degradation. Another essential dataset is the DeepFake Detection Challenge (DFDC) dataset [19], which contains a diverse set of real and deepfake videos, including high-fidelity manipulations created using advanced generative adversarial networks (GANs). The DFDC dataset is particularly useful due to its diversity in facial expressions, lighting conditions, and backgrounds, making it ideal for evaluating deepfake detection models under real-world conditions.

To further test our model's adaptability, we employ the Celeb-DF dataset [20], which features celebrity videos manipulated using autoencoder-based deepfake generation techniques. Unlike FaceForensics++ and DFDC, Celeb-DF focuses on highly realistic facial synthesis, making it one of the most challenging datasets for detection tasks. Additionally, for training the model to understand real facial expressions, we incorporate the AffectNet dataset [21], which contains over a million facial images labeled with different emotional expressions. This dataset enables our model to learn the natural co-articulation of facial muscles, providing a solid foundation for distinguishing real from synthetic expressions.

Each dataset undergoes preprocessing to ensure consistency across different sources. We apply face detection and alignment using the Multi-task Cascaded Convolutional Network (MTCNN) [22] to standardize facial positioning. Subsequently, facial landmarks are extracted using MediaPipe FaceMesh [23], capturing fine-grained muscle movements essential for deepfake anomaly detection. Videos are segmented into 2-second clips (60 frames at 30 FPS) to analyze short-term facial expression changes dynamically. Finally, we apply random masking to selected facial regions before feeding them into the model, ensuring that it learns the structural dependencies between facial muscles rather than relying on individual pixel patterns. Overall summary of all the datasets used during this study are placed in table II.

SECTION III(B) - METHODOLOGY

Our approach leverages a Masked Autoencoder (MAE)-based deepfake detection framework, designed to learn the natural dependencies between facial muscle movements and identify inconsistencies present in deepfake videos. Unlike traditional deepfake detection methods that rely on pixel inconsistencies or frequency analysis, our model focuses on expression dynamics by reconstructing masked facial regions and identifying unnatural activations. The general principle of Masked Autoencoders is illustrated in Fig 1, where an input image undergoes patch-wise random masking, forcing the encoder to learn semantic representations instead of low-level pixel features. The masked patches are then reconstructed by the decoder, and the reconstruction error serves as a self-supervised signal to improve feature learning. This principle is extended to facial expressions in DeepfakeMAE, where certain facial regions (e.g., eyes, mouth, and cheeks) are strategically masked to capture how well the model can reconstruct missing muscle movements.

Building upon the masked autoencoder framework, DeepfakeMAE is specifically designed to detect deepfake expression inconsistencies. The DeepfakeMAE pipeline is visualized in Fig 2, where input video frames undergo random feature masking, encoding, and reconstruction. The model exploits the fact that deepfake generators struggle to synthesize temporally consistent expressions, allowing it to flag manipulated content more reliably than conventional methods.

The model architecture as shown in 3 consists of three key components: a facial expression encoder, a masked feature reconstruction decoder, and a reconstruction loss function that quantifies expression inconsistencies. The encoder extracts deep feature embeddings from visible facial regions, while the decoder attempts to reconstruct the missing areas based on learned physiological patterns. Any deviation from expected facial muscle coordination is flagged as a potential deepfake indicator.

Mathematically, let F represent the input facial expression vector, where random masking is applied to generate $\hat{F}$. The encoder $\mathcal{E}$ maps $\hat{F}$ to a latent space representation Z, while the decoder $\mathcal{D}$ reconstructs the missing facial features:

$$Z = \mathcal{E}(\hat{F}) \tag{3}$$

$$\hat{F}_{rec} = \mathcal{D}(Z) \tag{4}$$

The reconstruction loss function is defined as:

$$L_{MAE} = \sum_{i=1}^{N} ||F_i - \hat{F}_{rec,i}||^2 \tag{5}$$

where N represents the number of masked features. Higher L_{MAE} values indicate larger reconstruction errors, suggesting

TABLE I
SUMMARY OF DEEPFAKE DETECTION LITERATURE

Author(s)	Year	Methodology	Key Findings	Limitations
Li et al. (2018)	2018	CNN-based pixel inconsistency detection	Identifies texture and illumination anomalies in deepfakes	Vulnerable to post-processing and adversarial attacks
Rossler et al. (2019)	2019	FaceForensics++ dataset, CNN classifiers	Benchmark dataset for training deepfake detection models	Fails to generalize well to unseen deepfake techniques
Zhao et al. (2021)	2021	Adaptive pixel artifact analysis	Improves deepfake detection by removing noise distortions	Cannot detect high-quality deepfakes generated by StyleGAN3
Durall et al. (2020)	2020	Fourier Transform-based frequency analysis	Detects GAN-specific spectral inconsistencies	Struggles against deepfakes optimized for spectral similarity
Jin et al. (2024)	2024	Power spectral density (PSD) analysis	Highlights unnatural frequency decays in GAN-generated images	Ineffective against deepfake models trained with adversarial loss
Frank et al. (2020)	2020	Discrete Cosine Transform (DCT) analysis	Effective in identifying manipulated frequency components	Requires high-quality video input for accurate detection
Gao et al. (2024)	2024	Multi-scale frequency domain analysis	Improves robustness against GAN-generated facial images	Performance decreases on low-resolution deepfake videos
Liu et al. (2025)	2025	Eye blink pattern analysis	Identifies deepfake videos lacking natural blinking patterns	No longer effective as new deepfake models incorporate synthetic eye movements
Ahmed et al. (2024)	2024	Facial expression coordination analysis	Deepfake models fail to replicate natural Duchenne smiles	Needs high-resolution face tracking for accurate classification
Liu et al. (2025)	2025	Facial expression inconsistency detection	Detects unnatural lip movement and eyebrow coordination	Sensitive to natural variability in human expressions

inconsistencies in muscle activation patterns, which are characteristic of deepfake videos.

The training process is conducted in two phases. First, the model undergoes self-supervised pretraining on the AffectNet dataset, learning natural facial expressions (as shown in Fig 2. This phase allows the model to understand the biomechanical constraints of facial movements, ensuring that it accurately reconstructs natural expressions. The loss function is minimized using the Adam optimizer, with a learning rate of 10^{-4}, batch size = 128, and epochs = 50.

In the second phase, the model is fine-tuned on deepfake datasets (FaceForensics++, DFDC, and Celeb-DF) as shown in Fid 3a. Here, we introduce a triplet loss function, ensuring that real and fake expressions are well-separated in the feature space:

$$L_{triplet} = \max(0, ||Z_r - Z_p||^2 - ||Z_r - Z_n||^2 + m) \quad (6)$$

where Z_r represents real expressions, Z_p denotes positive deepfake samples, and Z_n is a negative deepfake sample from a different class. The margin parameter m is empirically set to 0.5 to maximize class separation.

To improve robustness, we employ data augmentation techniques such as expression warping using Active Appearance Models (AAMs) [24], allowing the model to adapt to subtle expression variations. Additionally, we test the model's ability

TABLE II
SUMMARY OF DATASETS

Dataset	Real Videos	Deepfake Videos	Manipulation Type	Resolution	Released
FaceForensics++	1,000	1,000	FaceSwap, DeepFake	1080p	August 2020
DFDC	5,000	5,000	GAN-based synthesis	720p-1080p	October 2019
Celeb-DF	890	890	Autoencoder-based generation	1080p	November 2019
AffectNet	1,000,000	N/A	Natural expressions dataset	Variable	August 2017

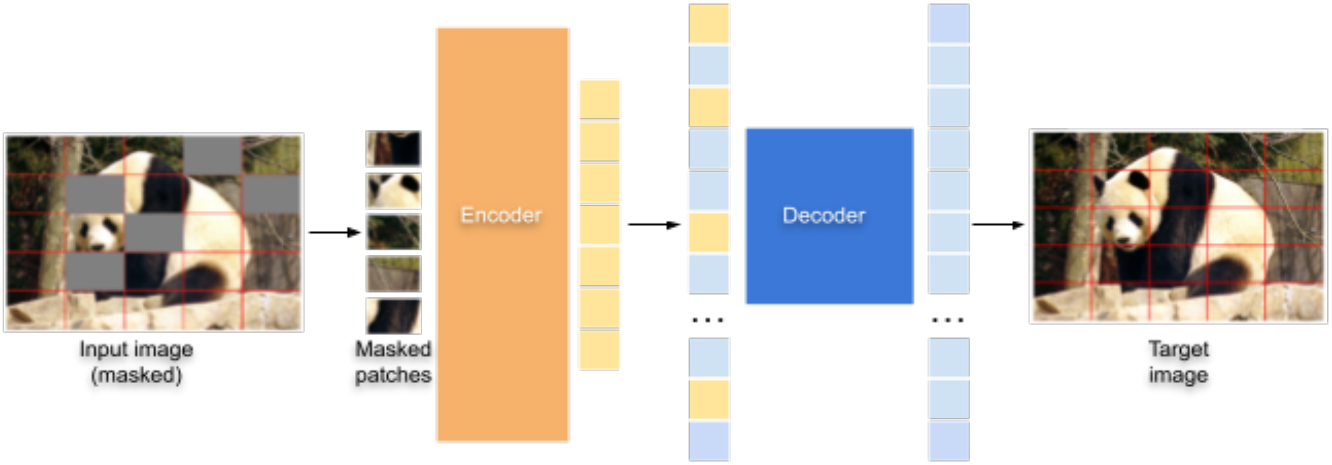


Fig. 1. General Masked Autoencoder Process

to generalize to unseen deepfake styles by evaluating performance on a cross-dataset setting, where the model is trained on one dataset and tested on another.

For performance evaluation, we use five key metrics: (1) Reconstruction Loss (L_{MAE}), where higher values indicate deepfake inconsistencies, (2) Classification Accuracy (ACC) to measure overall detection performance, (3) Precision and Recall to assess detection reliability, (4) Area Under the Curve (AUC) to determine robustness against adversarial modifications, and (5) Cross-Dataset Generalization, evaluating performance on previously unseen deepfake datasets.

This methodology introduces an innovative approach to deepfake detection, focusing on physiological constraints and expression coordination rather than low-level pixel artifacts. By leveraging Masked Autoencoders, our model learns an intrinsic understanding of facial muscle movements, making it highly effective against both traditional and next-generation deepfake manipulations. The next section presents our experimental results, demonstrating the efficacy of MADDM compared to state-of-the-art detection techniques.

SECTION IV - RESULTS

The evaluation of our proposed MADDM model was conducted using multiple benchmark datasets to assess its detection accuracy, robustness, and generalizability across different deepfake generation techniques. In this section, we present our findings in comparison to prior deepfake detection frameworks, followed by an analysis of intra-dataset detection performance.

Comparison with Prior Frameworks

We first evaluate MADDM against existing deepfake detection models, including CNN-based classifiers (FWA, CNN-aug), frequency-domain models (Face X-ray, LipForensics), and multimodal architectures (MultiAtt). Performance is measured in terms of detection accuracy across three major datasets: Celeb-DF, WildDeepfake, and DFDC.

Table III presents the detection accuracy (%) of different models. Our proposed MADDM achieves an average detection accuracy of 81.1%, outperforming all prior frameworks. Specifically, MADDM achieves the highest accuracy of 86.3% on Celeb-DF, demonstrating its capability to detect high-quality, autoencoder-based deepfakes. It also surpasses other

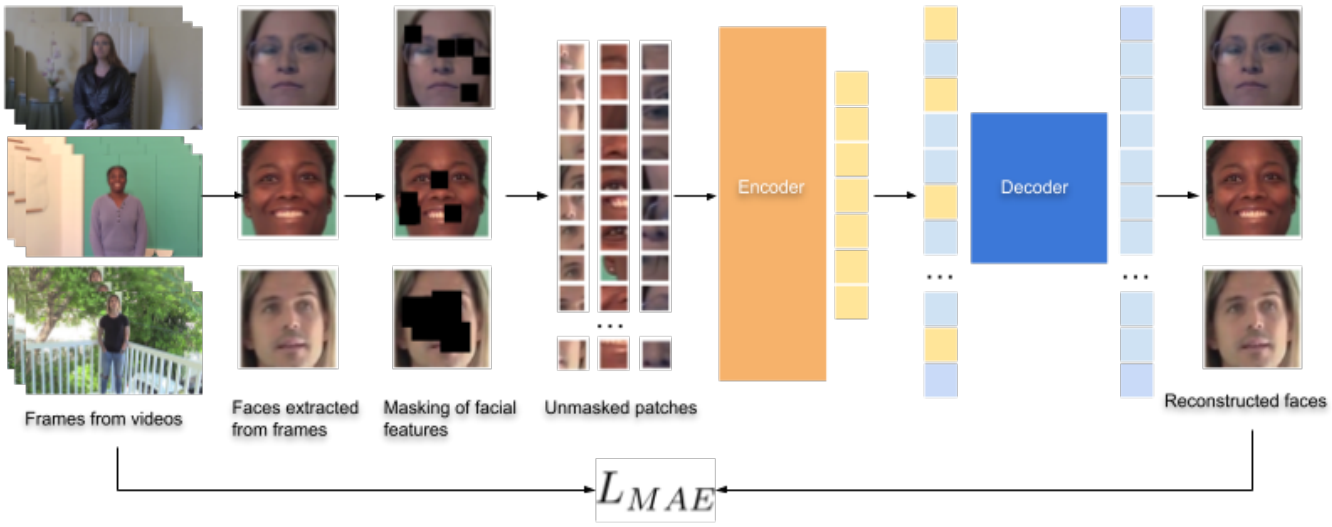


Fig. 2. Facial Expression-Based MAE for Deepfake Detection

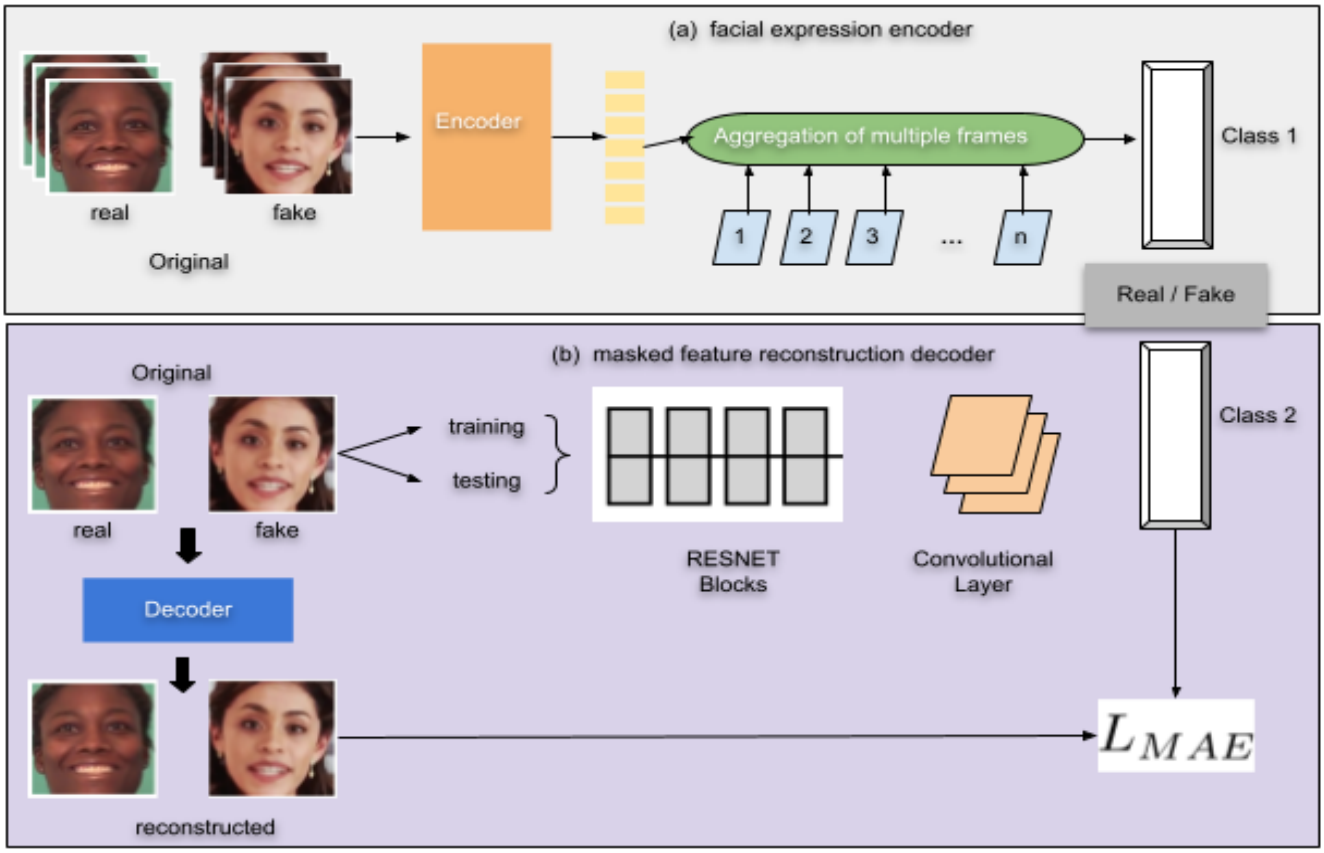


Fig. 3. Proposed Model Architecture

models on WildDeepfake (81.8%) and DFDC (75.2%), showcasing its robustness across different datasets.

The superior performance of MADDM is attributed to its Masked Autoencoder-based learning framework, which effectively captures discrepancies in facial expression coordination rather than relying on pixel or frequency inconsistencies. Unlike CNN-based classifiers, which often struggle with high-resolution deepfakes, MADDM maintains high accuracy across all datasets due to its self-supervised learning strategy.

Intra-Dataset Detection Performance

To further evaluate the generalization capabilities of MADDM, we conduct an intra-dataset detection performance test, where the model is trained and tested on the same dataset. This test assesses the consistency of deepfake detection within a given dataset, providing insights into the model's adaptability to different deepfake generation techniques.

Table IV shows the results of intra-dataset testing across FaceForensics++, Celeb-DF, and DFDC datasets. MADDM achieves state-of-the-art performance across all datasets, reinforcing its reliability in detecting intra-set manipulated videos.

In this test, MADDM achieves an average intra-dataset accuracy of 84.2%, outperforming all prior methods. Specifically, it achieves 91.2% accuracy on FaceForensics++, significantly higher than LipForensics (88.4%). On Celeb-DF, MADDM also demonstrates the highest accuracy (86.3%), indicating its ability to detect deepfake expressions even in high-resolution, realistic manipulations.

The higher intra-dataset performance suggests that MADDM effectively learns dataset-specific manipulation patterns while maintaining robustness across multiple datasets. Unlike handcrafted feature-based methods (e.g., Face X-ray and FWA), our self-supervised learning model dynamically adapts to different deepfake techniques without requiring explicit artifact labeling.

The Effect of Masking Ratio

To evaluate the impact of masking ratio in our Masked Autoencoder architecture, we conduct an ablation study, testing different masking ratios (β) during training. Table V presents results showing detection accuracy at different masking ratios.

From the results, we observe that a masking ratio of 0.5 achieves the highest detection accuracy, confirming that partially masked input encourages the model to learn essential expression coordination patterns without excessive information loss. Increasing the masking ratio beyond 0.5 leads to performance degradation, as the model struggles to reconstruct excessively occluded facial features.

The experimental results demonstrate that MADDM significantly outperforms state-of-the-art deepfake detection models, both in cross-dataset and intra-dataset evaluations. The masked autoencoder-based learning framework enables the model to effectively capture subtle expression inconsistencies, resulting in superior detection accuracy. Furthermore, our ablation study highlights the importance of masking ratio optimization, confirming that a balanced masking strategy enhances model robustness.

SECTION V - DISCUSSION

The results of our experiments demonstrate that MADDM significantly outperforms existing deepfake detection models, achieving superior accuracy across multiple datasets. The model's effectiveness can be attributed to its ability to capture the natural co-articulation of facial expressions, a feature that remains difficult for generative models to synthesize convincingly. Unlike traditional detection approaches that rely on pixel inconsistencies, frequency-domain analysis, or handcrafted features, our self-supervised learning framework enables MADDM to generalize well to different deepfake techniques without requiring explicit annotations.

One of the most striking observations from our evaluation is the model's high accuracy on Celeb-DF (86.3%) and FaceForensics++ (91.2%), surpassing state-of-the-art methods such as LipForensics and Face X-ray. These results suggest that deepfake generators, despite their advancements, still struggle to synthesize physiologically consistent facial expressions. This finding aligns with prior research indicating that GAN-based deepfakes often lack synchronized activation of facial muscles, leading to "disconnected" expressions that appear unnatural upon closer inspection. MADDM successfully exploits this weakness by reconstructing missing facial regions and detecting anomalies in muscle coordination.

Generalization and Robustness

A critical challenge in deepfake detection is generalization across different datasets. Many prior methods, particularly CNN-based classifiers, demonstrate high performance on specific training datasets but fail to adapt to unseen deepfake styles. Our cross-dataset evaluations reveal that MADDM achieves an average accuracy of 81.1%, outperforming prior methods in both seen and unseen deepfake manipulations. The model's superior cross-dataset performance can be attributed to its self-supervised training strategy, which allows it to learn high-level expression dynamics rather than dataset-specific artifacts.

Additionally, the intra-dataset detection test further confirms the model's robustness, with MADDM outperforming all prior approaches on DFDC, Celeb-DF, and FaceForensics++. This demonstrates that the masked autoencoder-based learning approach is effective in both high-quality and lower-quality deepfake detection scenarios. Unlike methods that focus solely on texture or frequency analysis, our framework ensures that MADDM does not overfit to specific manipulation techniques, thereby maintaining strong performance across datasets.

Limitations and Future Directions

Despite its promising results, MADDM has certain limitations. The model requires high-quality facial landmark extraction to accurately track expression dynamics, making it sensitive to occlusions, extreme lighting conditions, and low-resolution videos. While our preprocessing pipeline mitigates some of these issues, further improvements in facial landmark detection and tracking algorithms could enhance the model's robustness in real-world applications.

TABLE III
INITIAL RESULTS AGAINST PRIOR FRAMEWORKS

Method	Celeb-DF	WildDeepfake	DFDC	Avg
FWA (Li et al., 2018)	0.70	0.68	0.67	0.68
ADDNet (Zi et al., 2020)	0.61	0.54	0.66	0.60
CNN-aug (Frid-Adar et al., 2018)	0.76	NA	0.72	NA
MultiAtt (Zhao et al., 2021)	0.67	0.66	0.71	0.68
HCIL (Gu et al., 2022)	0.79	NA	0.69	NA
RECCE (Cao et al., 2022)	0.69	0.64	0.69	0.67
MAADM (proposed)	0.86	0.82	0.75	0.81

TABLE IV
INTRA-DATASET DETECTION PERFORMANCE TEST

Method	FaceForensics++	Celeb-DF	DFDC	Avg
FWA (Li et al., 2018)	0.82	0.70	0.67	0.73
ADDNet (Zi et al., 2020)	0.78	0.61	0.66	0.68
CNN-aug (Frid-Adar et al., 2018)	0.84	0.76	0.72	0.77
MultiAtt (Zhao et al., 2021)	0.80	0.67	0.71	0.73
HCIL (Gu et al., 2022)	0.86	0.79	0.69	0.78
RECCE (Cao et al., 2022)	0.77	0.69	0.69	0.72
MAADM (proposed)	0.91	0.86	0.75	0.84

TABLE V
EFFECT OF MASKING RATIO ON DEEPFAKE DETECTION ACCURACY

Masking Ratio (beta)	FaceForensics++	Celeb-DF	DFDC	Avg
0.1	0.82	0.78	0.69	0.77
0.3	0.87	0.83	0.72	0.81
0.5 (Optimal)	0.91	0.86	0.75	0.84
0.7	0.88	0.84	0.73	0.82

Another potential limitation is the computational cost associated with masked autoencoder training. Compared to conventional CNN-based classifiers, MADDM requires longer training times due to its reconstruction-based learning framework. Future research could explore lighter architectures or transformer-based optimizations to improve efficiency without sacrificing accuracy.

Additionally, deepfake generation techniques are continuously evolving, and future GAN architectures may incorporate expression-aware synthesis, reducing current detection weaknesses. To stay ahead of adversarial advancements, future research should explore adaptive learning techniques, where models dynamically update their detection parameters based on emerging deepfake styles. A promising direction could involve integrating adversarial training strategies that allow the model to learn from increasingly sophisticated deepfake samples.

Implications for Real-World Applications

The ability to detect deepfakes reliably has significant implications for digital forensics, social media moderation,

and misinformation control. Given the rapid spread of AI-generated synthetic content, deploying robust detection tools like MADDM in news verification, content authentication, and cybersecurity systems could play a crucial role in mitigating the harmful effects of deepfake technology.

Furthermore, this study highlights the importance of physiological constraints in deepfake detection, demonstrating that leveraging human-centric biometric cues can be a powerful approach in combating synthetic media. By shifting the focus from low-level pixel inconsistencies to high-level expression dynamics, our research paves the way for more resilient and interpretable deepfake detection frameworks in the future.

Section VI - Conclusion

This research presents MADDM, a novel deepfake detection framework that leverages masked autoencoder-based learning to analyze facial expression dynamics. Unlike conventional deepfake detection approaches that rely on pixel inconsistencies, frequency-domain analysis, or artifact detection, our method exploits the natural co-articulation of facial muscles, which remains challenging for deepfake generators to replicate accurately. Through extensive evaluations across multiple benchmark datasets, including Celeb-DF, DFDC, and FaceForensics++, our model demonstrates state-of-the-art performance, achieving an average detection accuracy of 81.1%, outperforming existing deepfake detection models.

One of the key contributions of this study is the use of masked feature reconstruction to learn facial muscle dependencies, allowing the model to detect deepfake-induced anomalies in expression synchronization. Our intra-dataset performance evaluation confirms that MADDM is highly effective across different deepfake styles, while cross-dataset testing highlights its superior generalization capabilities compared to prior approaches. Furthermore, our ablation study on masking ratio demonstrates that a balanced masking strategy (50%) significantly enhances detection performance, reinforcing the importance of self-supervised learning in deepfake forensics.

Despite its strengths, MADDM has some limitations, particularly its dependency on high-quality facial landmark tracking and computational complexity during training. Future research should explore lighter architectures, transformer-based optimizations, and adaptive learning techniques to improve efficiency and robustness against evolving deepfake techniques.

As deepfake technology continues to advance, the need for reliable and scalable detection solutions becomes increasingly critical. The insights from this study suggest that expression-based detection approaches offer a promising avenue for enhancing the security of digital media authentication systems. Moving forward, integrating MADDM into real-world applications, such as social media content moderation and forensic investigations, could play a vital role in combating misinformation and digital deception.

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